

Temperature Sensor Arduino Project

A temperature sensor is an electronic component that measures temperature and sends the value to a controller like an Arduino. In your code, the temperature sensor is connected to A0 (Analog Pin 0) of the Arduino.

Common sensors used:

LM35 → Gives output directly in °C

TMP36

DHT11/DHT22 (digital sensors)

Your code looks similar to using an LM35/TMP36-type analog temperature sensor.

Explanation of the Code

```
int temp = 0;
```

Creates a variable named temp

Stores the temperature value

setup() Function

```
void setup()
```

```
{
```

```
pinMode(A0, INPUT);
```

Sets pin A0 as input

Arduino reads sensor data from this pin

```
Serial.begin(9600);
```

Starts serial communication

Used to display temperature in Serial Monitor

9600 = communication speed (baud rate)

```
pinMode(12, OUTPUT);
```

```
pinMode(11, OUTPUT);
```

```
pinMode(10, OUTPUT);
```

Pins 12, 11, and 10 are outputs

Usually connected to LEDs or devices

Example:

Pin 12 → Green LED

Pin 11 → Yellow LED

Pin 10 → Red LED

loop() Function

This runs continuously.

Reading Temperature Sensor

```
temp = map(((analogRead(A0) - 20) * 3.043), 0, 1023, -20, 120);
```

Step-by-step:

1. `analogRead(A0)`

Reads analog voltage from sensor.

Arduino ADC range:

0V → 0

5V → 1023

Example:

```
analogRead(A0) = 150
```

2. `(analogRead(A0) - 20)`

Adjusts sensor offset.

3. `* 3.043`

Scales the value.

4. `map()`

Converts sensor value into temperature range.

```
map(value, 0, 1023, -20, 120)
```

Meaning:

0 sensor value → -20°C

1023 sensor value → 120°C

So finally:

temp

contains temperature in Celsius.

Printing Temperature

```
Serial.println(temp);
```

Displays temperature in Serial Monitor.

Example output:

28

29

30

LED Control Logic

First LED

```
if (temp < 30) {
```

```
digitalWrite(12, HIGH);
```

```
} else {
```

```
digitalWrite(12, LOW);
```

```
}
```

If temperature is below 30°C:

LED on pin 12 turns ON

Otherwise:

OFF

Second LED

```
if (temp < 60) {
```

```
digitalWrite(11, HIGH);
```

```
}
```

If temperature is below 60°C:

LED on pin 11 ON

Third LED

```
if (temp < 90) {
```

```
digitalWrite(10, HIGH);
```

```
}
```

If temperature is below 90°C:

LED on pin 10 ON

Delay

```
delay(10);
```

Waits 10 milliseconds before next reading.

Helps:

Stable readings

Better simulation performance

Working Summary

Temperature Range vs LEDs

Simple Working Principle

Temperature sensor senses heat

Arduino reads analog voltage

Code converts voltage into temperature

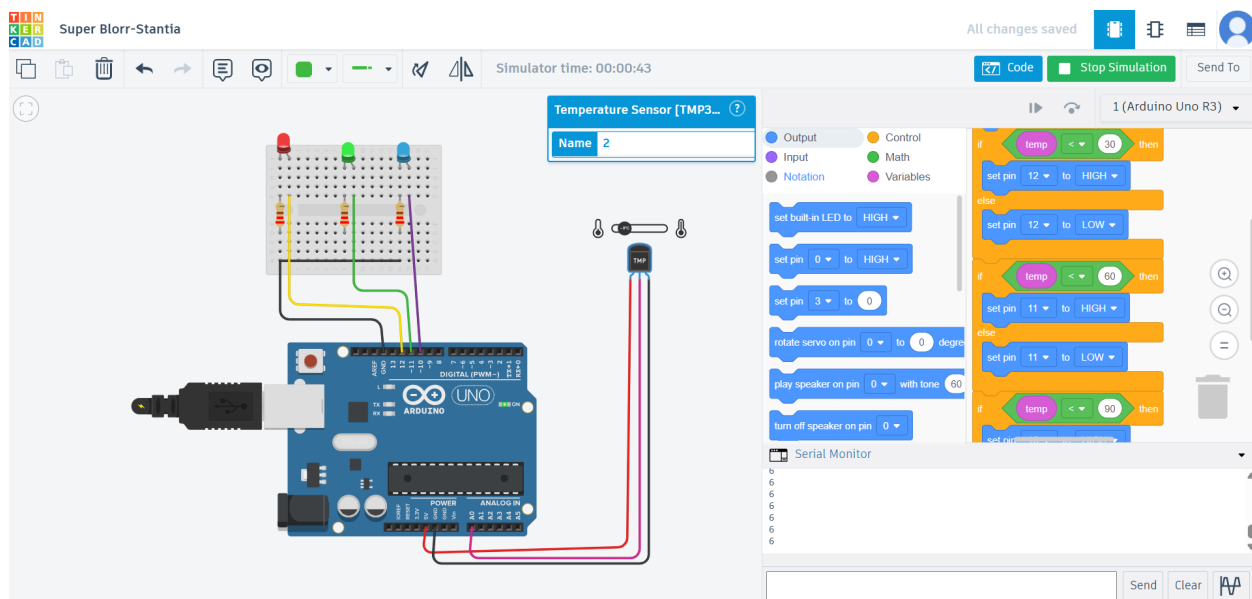
Temperature shown in Serial Monitor

LEDs turn ON/OFF based on temperature level

Sensor Connection Example

LM35 Pin Connections

Tinkercad Circuit Simulation



Arduino Code Screenshot

```
// C++ code
//
int temp = 0;

void setup()
{
  pinMode(A0, INPUT);
  Serial.begin(9600);
  pinMode(12, OUTPUT);
  pinMode(11, OUTPUT);
  pinMode(10, OUTPUT);
}

void loop()
{
  temp = map(((analogRead(A0) - 20) * 3.043), 0, 1023, -20, 120);
  Serial.println(temp);
  if (temp < 30) {
    digitalWrite(12, HIGH);
  } else {
    digitalWrite(12, LOW);
  }
  if (temp < 60) {
    digitalWrite(11, HIGH);
  } else {
    digitalWrite(11, LOW);
  }
  if (temp < 90) {
    digitalWrite(10, HIGH);
  } else {
    digitalWrite(10, LOW);
  }
  delay(10); // Delay a little bit to improve simulation performance
}
```